

BREATHBEAT: Individual Progress Report

1. Project Overview & Research Hypothesis

BREATHBEAT is an interactive web platform designed to test whether sonifying 4-2-6 diaphragmatic breathing (4s inhale, 2s hold, 6s exhale) with Indian classical music helps people recover from mental fatigue faster and with less effort than voice counting or silent rest.

We hypothesized that replacing spoken counts with the continuous melodic glides (*Meend*) of **Raga Bhupali** over a continuous **Tanpura drone** provides an intuitive musical guide. As the melody glides upward (Arohana: Sa to Sa'), it naturally cues inhalation. When it holds on the resting note (*Ga*), it anchors breath retention (*Kumbhaka*). As it gently glides back down (Avarohana: Sa' to Sa), it guides a relaxed exhalation. This lets listeners keep rhythm naturally without having to count numbers in their heads.

2. Individual Progress

For Sprint 3, I was responsible for developing the full evaluation website and integrating its core functional modules:

1. Guided 5-Stage Testing Flow:

- Implemented the complete experimental sequence: a baseline Stroop check to record fresh focus, a 3-minute mental math test to exhaust working memory, a post-fatigue Stroop check to measure the exact drop in reaction speed, a 5-minute guided 4-2-6 breathing break across three randomized study arms, a recovery Stroop check to measure restored focus, and a post-task NASA-TLX workload survey.
- Built dual runtime options: a **Fast Evaluation Mode** (~2 minutes total for rapid grading and demonstration) and a **Full Clinical Protocol** (14 minutes).
- Automated the calculation of the Cognitive Recovery Score based on the formula from our proposal: $\text{Recovery (\%)} = [(\text{Fatigued Time} - \text{Recovered Time}) / (\text{Fatigued Time} - \text{Baseline Time})] * 100$.

2. Audio Synthesizer Engine (Functional Prototype):

- Built real-time procedural sound generation using the browser's Web Audio API to prevent audio stutter and ensure tight synchronization with breath transitions.
- Programmed a 4-string Tanpura drone with resonance filtering and scheduled smooth Raga Bhupali glides to the 4-2-6 cycle, synced to an animated lotus visualizer.
- The current synthesizer serves as a functional placeholder: it accurately executes the 4-2-6 timing and note transitions, but the synthetic timbre sounds somewhat rough and artificial rather than deeply meditative.

3. Analytics Dashboard & Pipeline Validation:

- Created a pre-programmed sample dataset of 20 participants across the three study groups to test and validate the data visualization and analysis pipeline.

- Programmed an automated one-way ANOVA calculator to test for statistical significance across group recovery rates once live participant data is collected.
 - Built interactive Chart.js graphs showing speed changes across all test stages, live session data merging, and single-click CSV and JSON data export.
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3. Sprint 4 Targets

1. **Source Authentic Meditative Audio:** Replace the current placeholder synthesizer sounds with authentic, high-quality acoustic recordings of a real Tanpura and traditional melodic instruments (such as the bansuri or sitar) that feel truly calming and natural for the breathing session.
2. **Website Deployment & Live Hosting:** Deploy the production web app to a public cloud platform (e.g., Vercel / GitHub Pages) so classmates can open and run the test conveniently on their own devices instead of running it locally.
3. **Controlled In-Person Trials:** Administer the full 14-minute protocol to our classmates in the Music Workshop studio, replacing the synthetic validation dataset with verified empirical experimental data.